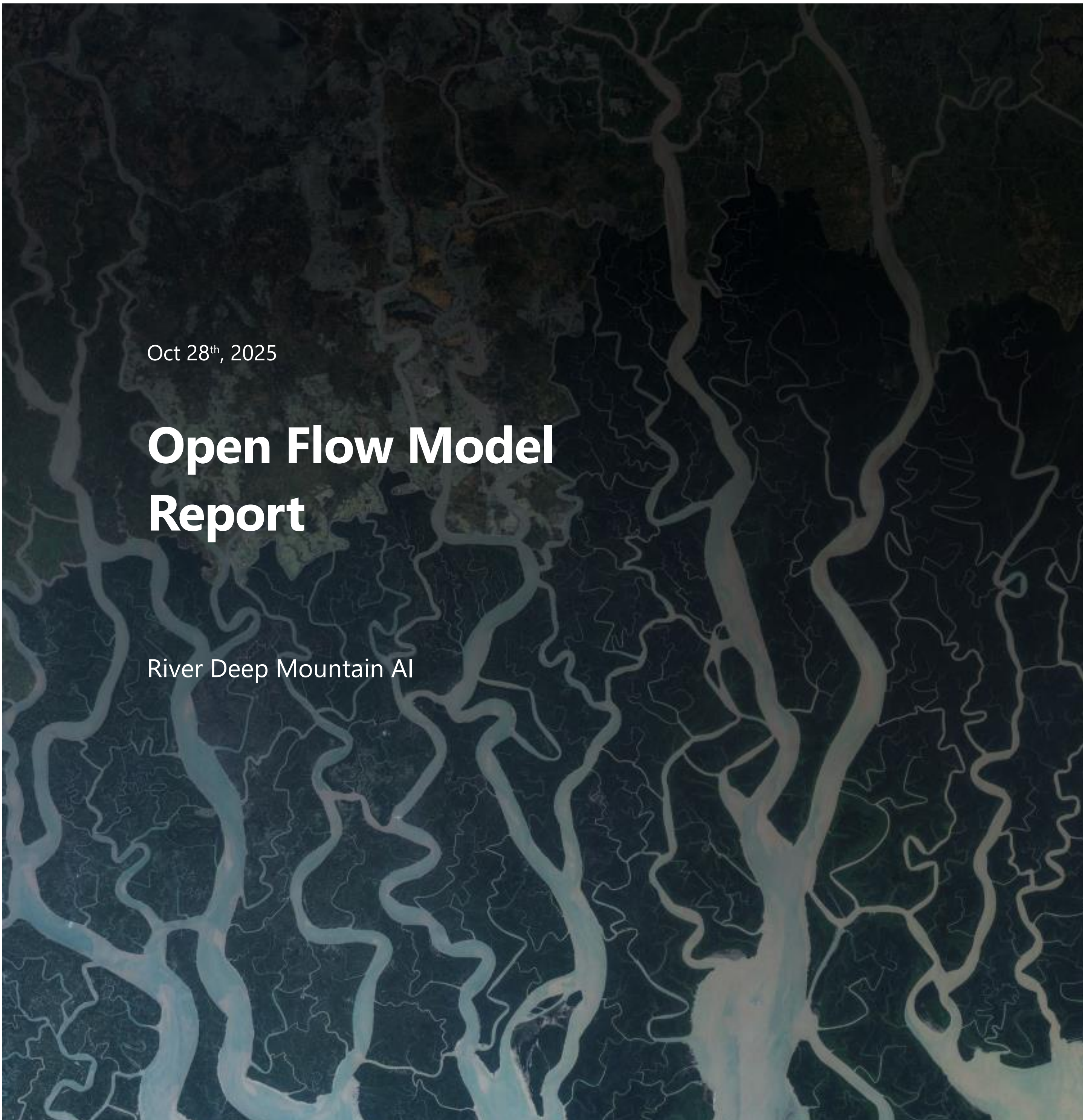


Oct 28th, 2025

Open Flow Model Report

River Deep Mountain AI

Table of Contents

1 Introduction

Understanding river flow is essential for effective water resource management. River flow data, typically measured in cubic meters per second (m^3/s), describes the volume of water moving through a waterbody at a given time. This information is critical for:

- Predicting floods
- Identifying and tracing pollution sources
- Managing ecosystems and water quality

However, gauged flow data are often sparse or entirely missing, especially in smaller or remote rivers. This is largely due to the high cost of installing and maintaining monitoring stations, leaving many catchments without direct observations. With a changing climate, understanding changes in flow is becoming increasingly important.

To address these data gaps, hydrologists often rely on surrogate data from comparable catchments, deterministic hydrology models, or conduct expensive field campaigns. These approaches, while useful, are limited in scalability and timeliness.

1.1 Objective

This project aims to estimate daily mean river flow (m^3/sec) in ungauged catchments using a machine learning approach that leverages both static and dynamic data sources.

Estimating daily mean flow (m^3/sec) is the main objective of our work; however, this metric can be used to infer other key flow metrics, such as mean flow and Q95 (flow exceeded or equalled 95% of the time; usually taken as a metric of low flow). During phase 3, we aim to evaluate the model's usefulness for estimating these metrics as well.

The model, and its associated performance metrics, presented in this report, represents the first and second iteration of our ML-model addressing the challenges mentioned above. Section 2 through 6 focus on the first iteration, whereas section 7 focuses on the second, and final, iteration and its associated performance.

1.2 Glossary

- **Catchments**: In this report we refer to the area contributing to the flow of a river at a flow gauging station as a catchment. Consequently, the catchment of flow gauging station a will encompass the catchment of flow gauging station b if b is upstream of a on the same waterbody. In the development of our model, we have used data from 409 flow gauging stations, representing 409 catchments. This adheres to the use of catchments in similar literature (Arsenault et al., 2019; Kratzert et al., 2019; Lane et al., 2019).

- **Static data**: Throughout the report, we distinguish between static and dynamic data. Static refers to catchment attributes that rarely change, or change over longer periods of time, such as topography, land cover, and soil.

- **Dynamic data**: Dynamic data refers to data that are changing daily, and mainly covers weather data such as rainfall, temperature, radiation, etc.

1.3 Non-stationarity

Due to the inherent non-stationarity of the challenges addressed in this project, it is important to continuously re-train and fine-tune the model to make sure it is up to date with the latest trends and variances of our climate and environment. The model was trained on publicly available data up to 2015. In the future, the model may not reflect updated scientific understandings, environmental conditions, or regulatory standards.

2 Data Sources

- **CAMELS-GB**: A comprehensive dataset of catchment attributes and flow data across Great Britain from 1970 to 2015.

- **Hydrology Data API**: Provides flow flags, flow and other hydrological indicators across England.

- **Chalk streams in England**: Dataset provided by Natural England. Identifies chalk streams, which are unique geological catchments.

- **Additional Variables**: To be integrated as they become available. Please note: a discussion is still going on regarding accuracy for some of these variables.

2.1 Feature selection

Minimal feature engineering is applied, aside from standard scaling. For attributes, notably, feature selection has taken into consideration "non-dependent" variables only (except for **Baseflow Index**, but just in some samples), which helps mitigate redundancy risks—a concern raised during earlier discussions. For time series, three additional fields have been added to provide the model with temporal context, both in terms of the period in the year and general time reference (see section 8.2). See appendix for a detailed overview of feature selection (sections 8.1 and 8.2).

BFI refers to the Baseflow Index. This is a hydrological metric that quantifies the proportion of streamflow that comes from groundwater discharge rather than surface runoff. It’s used to characterise catchment response and influences model selection and performance in river flow simulations

3 Model selection

The model architecture is structured into two branches:

1. Temporal Branch: A Long Short-Term Memory (LSTM) layer processes time-varying inputs such as weather conditions (Kratzert, et al., 2019; Arsenault et al., 2023).

2. Static Branch: An encoder, composed by 2 following dense layers (with reducing size) handles static catchment attributes (e.g., land use, soil type)

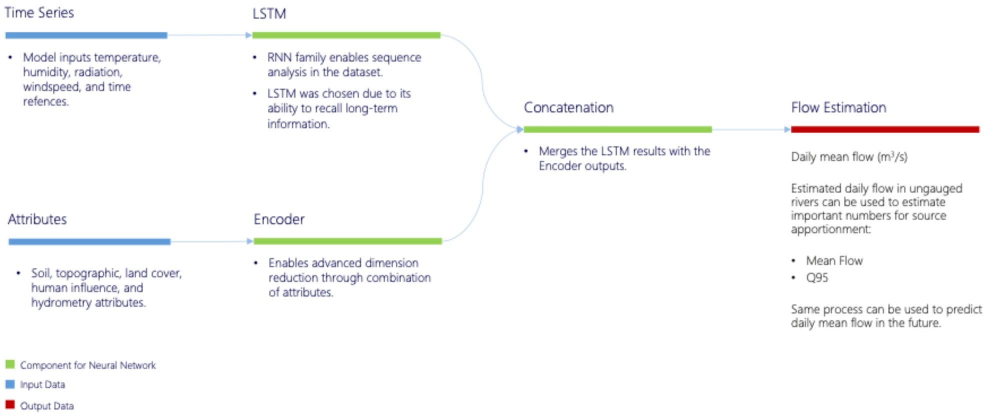


Fig. 1: Model architecture

The outputs of both branches are concatenated and passed to a dense layer, which produces a single flow prediction for each catchment temporally correspondent to the final step of each time window. The length of each time window was chosen to be 30 days, in agreement with Subject Matter Experts (SMEs). 30 days is a standard number used in traditional hydrological modelling and ensures that catchment wetness is considered. To ensure results are non-negative, the last dense layer has been set to have either an exponential or softplus activation function. Valuation of this option is part, similarly to other model parameters, of hyperparameter evaluation.

Please note that other Neural Network architectures, like inputting static variables to the LSTM, would have led to an extreme increase in required computing power (already quite demanding) without a real expectation of performance enhancement.

4 Training and testing strategy

To effectively train and evaluate the model, a structured approach tailored to the nature of the dataset has been adopted, which consists of time series data from multiple flow gauging stations and static attributes. Here's how the process works:

4.1 Per-catchment Train-Test Split

Instead of splitting the dataset per flow gauging station, each flow gauging station's time series has been split 70/30 individually into training and testing segments. This ensures that the model is tested on unseen future data for each flow gauging station, maintaining a realistic forecasting scenario. See diagram below:

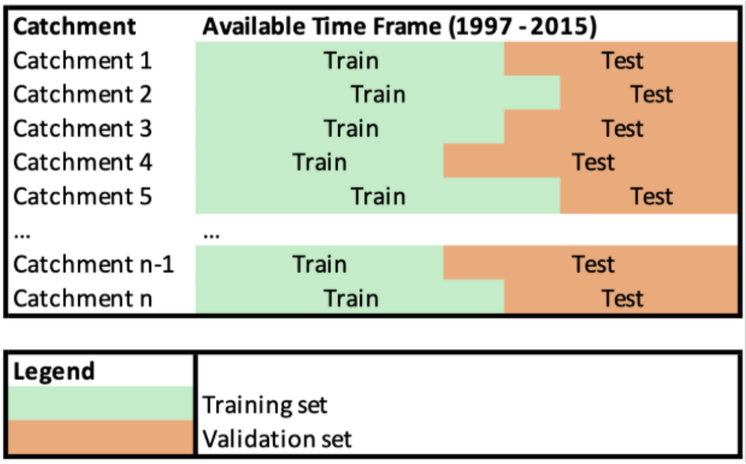


Fig. 2: Train-test split

4.1.1 Validation: Train-Validate Split

For the validation exercise we wanted to recreate the most realistic environment, we have split the data so that catchments do not span both train and validate. This is to ensure that there is no leakage and that the catchments in our validation set are unseen when training. To achieve this, all catchments were allocated with train or validate randomly as close to a 70% and 30% split as possible. This introduced other issues that we address in the validation section of this report. See diagram below:

Catchment	Available Time Frame (1997 - 2015)
Catchment 1	Train
Catchment 2	Validation
Catchment 3	Train
Catchment 4	Train
Catchment 5	Validation
...	...
Catchment n-1	Validation
Catchment n	Validation

Legend	
(green)	Training set
(orange)	Validation set

Fig. 3: Train-validation split

4.2 Time Windowing

The time series data is then segmented into 30-days sliding windows (sequences of fixed length). These windows serve as input samples for the model, allowing it to learn temporal patterns and antecedent effects such as soil moisture.

4.3 LSTM Input Preparation

The time windows are used to feed the LSTM (Long Short-Term Memory) network, which is well-suited for capturing dependencies in sequential data like weather or flow conditions.

4.5 Static Feature Integration

Alongside the dynamic time series data, each flow gauging station is associated with static attributes (e.g., catchment characteristics). These are encoded all together, passed to the encoder, and concatenated with the LSTM output, enriching the model with contextual information.

4.6 Model Training

The model is trained using the training portion of each flow gauging station's data. This includes both the time windows and the corresponding static features.

4.7 In-sample Model Evaluation

Finally, the model is evaluated on the test portion of each flow gauging station's data. This setup ensures that the model's performance reflects its ability to generalize to unseen time periods. The performance metrics from these in-sample evaluations is presented in section 6.

5 Evaluation metrics

To assess the performance of the model in estimating river flow across diverse catchments, a multi-faceted evaluation strategy has been developed, that captures both overall accuracy and catchment-specific behaviour.

5.1 Loss Functions and Metrics

As a regression problem, the model might be trained using:

- Mean Absolute Error (MAE)

This loss function is chosen based on the specific evaluation goals—MAE for absolute accuracy.

During training and evaluation, the following metrics are monitored regardless of the loss considered:

- MAE: Measures the average magnitude of errors in absolute terms
- MAPE: Captures the average percentage error, useful for understanding relative performance across varying flow magnitudes
- Root Mean Squared Error (RMSE): Highlights larger errors due to its quadratic nature, offering insight into outlier behaviour

- Nash-Sutcliffe Efficiency (NSE): A commonly used metric for hydrological models, assessing the predictive power and accuracy of a model (section 6.1.6).

5.2 Catchment-Level Aggregation

Given that the dataset includes time series from multiple flow gauging stations (each representing a different catchment), all metrics are also computed and reported as catchment-averaged values. The errors are also geographically mapped to identify any spatial patterns (see section 6.1.4 and 6.1.6).

5.3 Error Concentration Analysis

Beyond average metrics, an error concentration analysis has been conducted at the catchment level. This involves identifying:

- Which catchments consistently exhibit higher errors
- Whether errors are localized to specific hydrological or geographical conditions by looking for potential biases in model performance across different catchment types (e.g., chalk streams vs. others)

This layered evaluation approach provides a comprehensive understanding of model performance, both globally and locally, and helps guide future improvements in model generalisation and fairness. In addition, visualization through maps helps with geographical reference.

6 Performance metrics

The table below presents results from a series of experiments conducted using slightly different datasets. While the datasets vary marginally, the loss function and the structure of the final layer have been kept consistent across all experiments to ensure comparability, at least now.

Table 1: Overview of the performance across different experiments.

exp.	bfi	cs	csf	final activation func.	label_field	loss	obs_test_MAE	obs_test_MAPE	obs_train_MAE	obs_train_MAPE
1	Y	Y	N	exponential	discharge_vol	MAE	0.911	232.332	0.75	285.983
2	N	Y	Y	exponential	discharge_vol	MAE	0.931	273.941	0.785	372.154
3	Y	Y	Y	exponential	discharge_vol	MAE	0.932	233.481	0.81	293.171
4	N	Y	N	exponential	discharge_vol	MAE	0.978	313.554	0.846	402.277
5	Y	N	N	exponential	discharge_vol	MAE	1.15	28.799	0.992	53.523
6	N	N	N	exponential	discharge_vol	MAE	1.185	43.929	1.039	71.175

Table 1: Performance metrics

For clarity:

- bfi: inclusion bfi as variables, attributes dataset only
- cs: presence of chalk streams, both attributes and time series datasets
- csf: presence of chalk streams flag (a Boolean which specify whether a catchment is in a chalk stream), attributes dataset only
- dense_output_activation: activation function for the last dense layer
- label_field: which label used, (plain or with logarithmic transformation)
- fields for model performance have “obs”, standing for "observations" as all the time windows are here considered (without catchment grouping)

The following graphs in section 6 are based on the "best model" above (first row). MAE is used as a ranking tool in this case, as it was used a loss function for training the models, which makes it the fairest way to rank the models against each other. For future iterations, MAPE or other metrics can be used as the loss function to evaluate which model is the best performing.

6.1 Performance analysis

It is not possible to provide the total number of graphs produced so far, so only those with a general focus will be shared, along with a selection of tables that highlight specific aspects.

6.1.1 Observation-level performance

Due to the high number of observations and problems with scale, graphs may not be the best option for obtaining insights into the model results. For this reason, a table for each set (train and test) has been produced (see table 2 and table 3 below).

TRAIN	Actual flow	Predicted flow	Difference	Absolute difference	% Differences	Absolute % differences
mean	5,87	5,679	-0,191	0,746	285,821	285,983
std	20,223	19,369	2,652	2,552	8496,818	8496,813
min	0	0,004	-216,546	0	-0,982	0
1%	0,011	0,048	-8,678	0,002	-0,554	0,003
5%	0,056	0,093	-1,944	0,011	-0,365	0,015
10%	0,108	0,144	-0,796	0,021	-0,27	0,029
15%	0,167	0,201	-0,433	0,031	-0,21	0,044
20%	0,235	0,262	-0,263	0,042	-0,165	0,06
25%	0,312	0,33	-0,165	0,054	-0,127	0,075
30%	0,4	0,409	-0,103	0,067	-0,094	0,092
50%	0,96	0,95	0,016	0,152	0,024	0,171
70%	2,536	2,498	0,101	0,371	0,179	0,295
75%	3,417	3,346	0,143	0,484	0,242	0,344
80%	4,79	4,688	0,208	0,656	0,327	0,41
85%	7,103	6,919	0,321	0,943	0,454	0,508
90%	11,669	11,326	0,546	1,51	0,682	0,697
95%	25,103	24,065	1,167	3,069	1,308	1,308
99%	89,038	85,768	4,481	10,52	5,809	5,809
max	876,836	791,054	131,634	216,546	1242464,377	1242464,377
skew	9,837	9,866	-8,494	13,149	47,023	47,023
kurt	142,002	143,667	292,694	328,818	3225,88	3225,885
range	876,836	791,05	348,18	216,546	1242465,359	1242464,377

Table 2: Distribution of training dataset and performance. Metrics are across 2,403,628 readings of river flow.

Overall, actual and predicted values seem sharing the same distribution, although noticeable differences show up in the distribution tails.

In terms of residuals, statistics show a negative average value, but positive median, which means that, in terms of number of observations, the model tends to overestimate flow, but when it underestimates flow, the magnitude of underestimation exceeds that of overestimation.

That is also clear by looking at the values for the 1^{st} and the 99^{th} percentile. Finally, percentage difference (both with and without sign), show pronounced asymmetry behaviour in the tails, but on the opposite direction.

TEST	Actual flow	Predicted flow	Difference	Absolute difference	% Differences	Absolute % differences
mean	5,99	5,773	-0,217	0,911	232,165	232,333
std	21,311	20,076	3,578	3,466	7619,649	7619,644
min	0	0,005	-339,853	0	-0,981	0
1%	0,011	0,05	-10,439	0,002	-0,57	0,003
5%	0,058	0,096	-2,239	0,011	-0,379	0,016
10%	0,111	0,148	-0,902	0,022	-0,282	0,032
15%	0,173	0,205	-0,474	0,032	-0,221	0,048
20%	0,238	0,267	-0,274	0,044	-0,173	0,065
25%	0,313	0,335	-0,167	0,056	-0,133	0,082
30%	0,397	0,413	-0,102	0,069	-0,097	0,1
50%	0,948	0,946	0,019	0,16	0,031	0,185
70%	2,556	2,512	0,108	0,405	0,199	0,313
75%	3,459	3,383	0,154	0,537	0,264	0,363
80%	4,894	4,759	0,227	0,74	0,35	0,429
85%	7,295	7,068	0,352	1,086	0,477	0,528
90%	11,943	11,593	0,613	1,783	0,702	0,716
95%	25,36	24,419	1,388	3,717	1,294	1,294
99%	89,694	86,069	6,018	13,39	5,396	5,396
max	982,414	739,004	236,477	339,853	1044993,712	1044993,712
skew	10,732	10,402	-10,157	15,901	52,493	52,493
kurt	172,185	160,127	472,22	521,915	3691,202	3691,207
range	982,414	739	576,33	339,853	1044994,693	1044993,712

Table 3: Distribution of test dataset and performance. Metrics are across 940,116 readings of river flow.

Test set statistics show that, overall, model has a good grip on those time series that have not been used to train the model on, although along several dimensions, there is a slight downgrade of performance, which is expected.

In the following sections, we explore the performance of the model in detail.

6.1.2 Catchment-level performance

As part of investigating the performance of the model, we have looked at the cumulative error of the model across catchments. Looking at the graph, it becomes evident that there is a problem with a subset of catchments (around 5%) where a majority of the errors are concentrated. In the following, these 5% are referred to as the "poor performing catchments".

Below is a composite chart showing this:

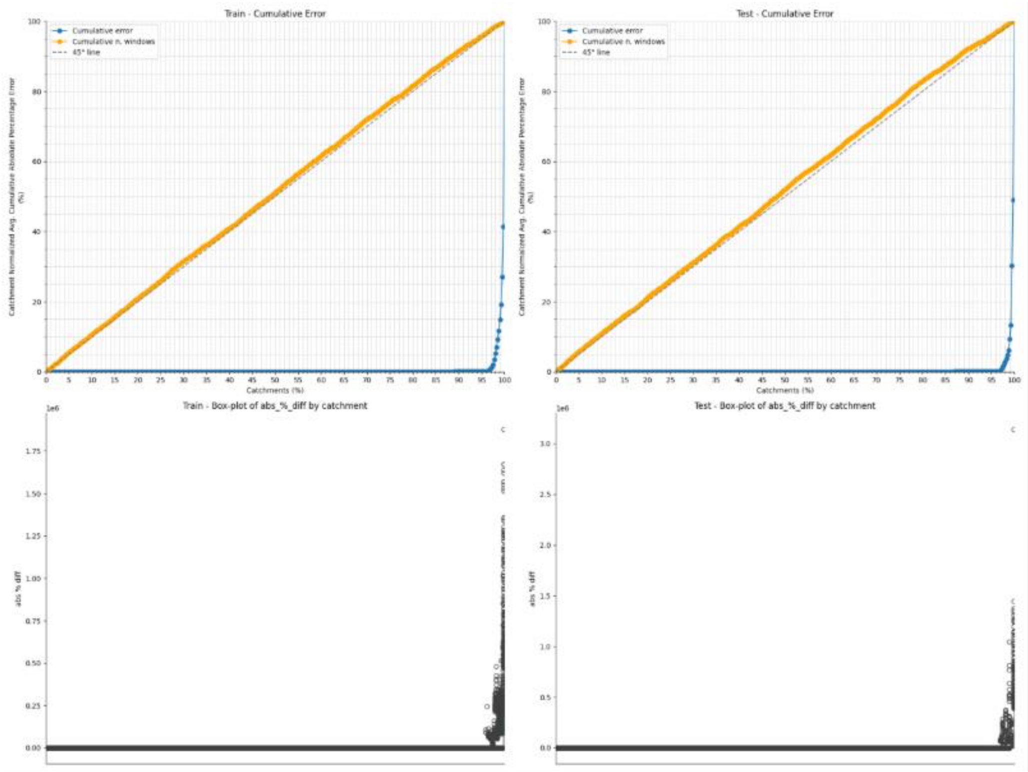


Fig. 4: Cumulative error and box plots

In the first row, the graphs show an explosion of errors for the very last set of catchments. The orange line represents the cumulative percentage of samples, and the blue line represents the cumulative percentage of errors. We would expect the orange and blue line to be similar, as it would indicate that each catchment has the same number of errors. However, what we see here is that a relatively small number of catchments (around 5%) is causing most of the errors.

In the second row, we see the distribution of the errors for the corresponding catchments (as plotted in the first row).

6.1.3 Investigation of poor performing catchments

Ahead of the first release of our flow model, we have investigated the "poor performing" catchments detailed above. However, so far, we have not been able to link the "poor performing" catchments to specific attributes. Below is

an example focused on catchment area:

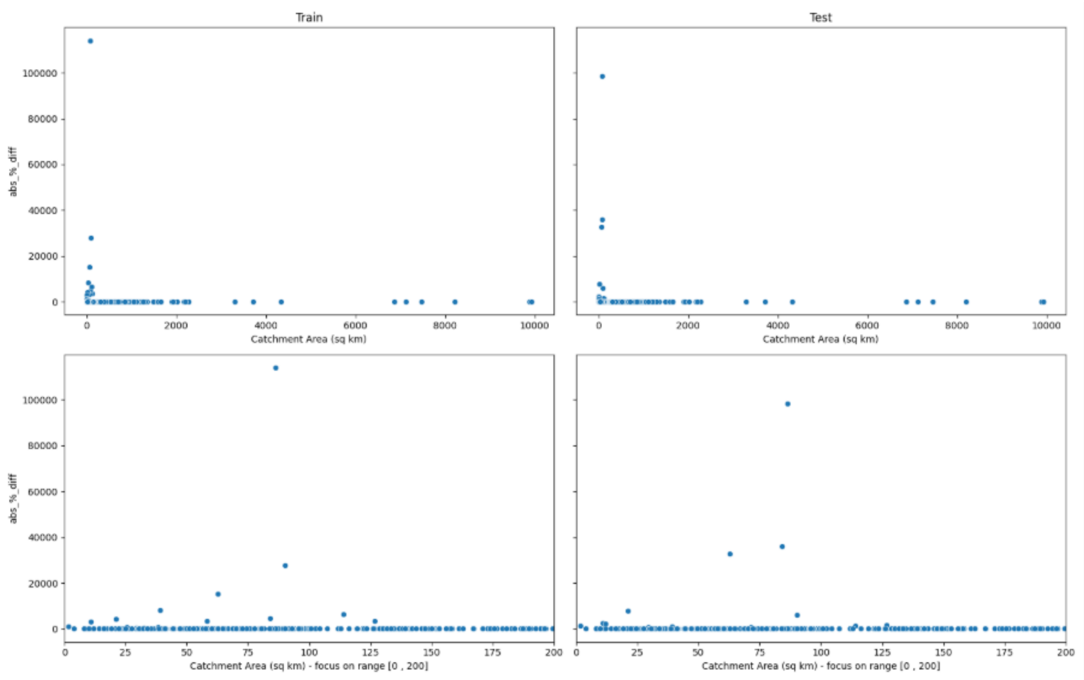


Fig. 5: Catchment area: These graphs show the relationship between catchment area and Mean Average Percentage Error. Although a few very big errors exist in smaller catchments, this alone is not enough to conclude that catchment area is the main reason for the high error.

As River Deep Mountain AI continues, we will continue to investigate the possible reasons for these catchments to be performing poorly. So far, we have investigated the link between poor performing catchments and known chalk streams, which has shown some relationship, although it does not account for all poor performing catchments (see section 6.1.4).

6.1.4 Geographical representation of performance

Below the localization of the poor and good performing catchments across England. We have generated three maps, focusing on Mean Error (a), MAE (b), MAPE (c) and MAPE excluding chalk streams (d).

a. Focus on mean error:

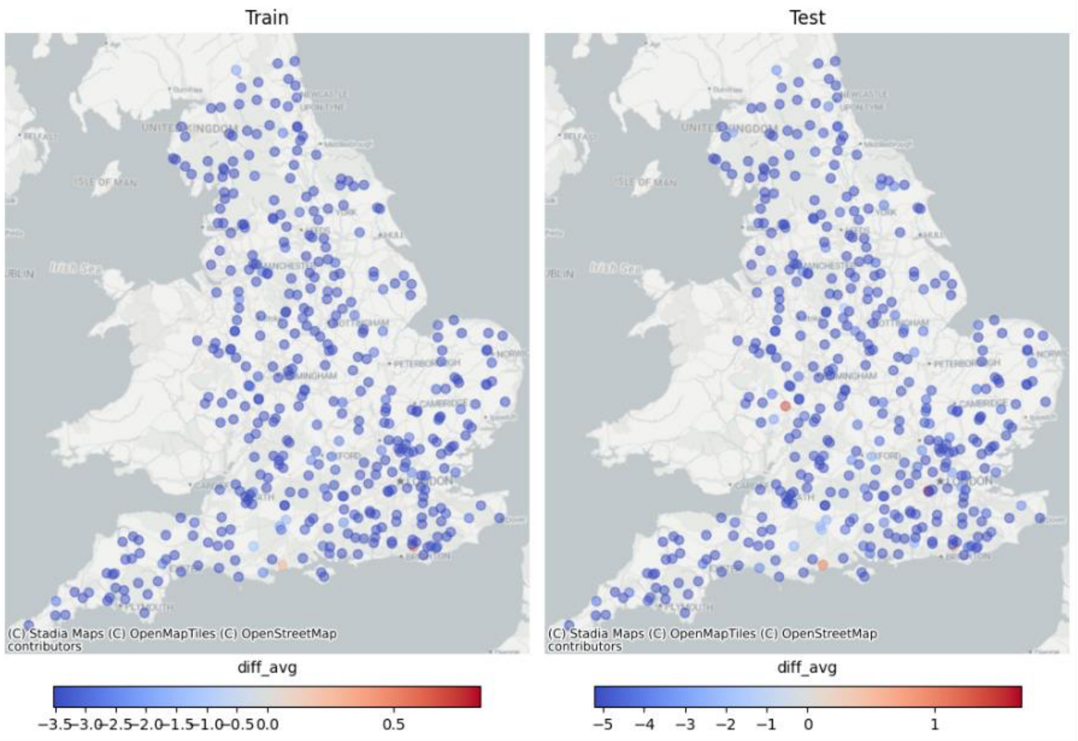


Fig. 6: **Average error:** This map shows the distribution of mean error across 409 catchments. This map shows that our model currently under-estimates the daily mean flow in most of the catchments.

b) Focus on average MAE:

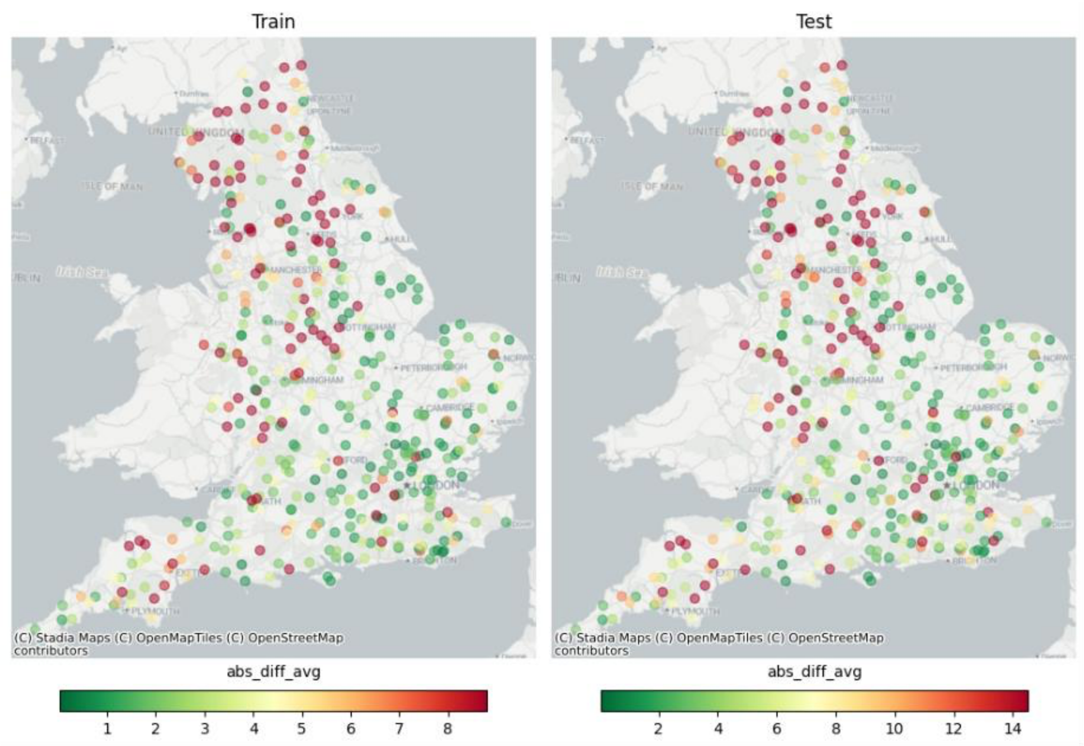


Fig. 7: Average MAE: This map shows the distribution of mean absolute error across 409 catchments. It's evident that larger catchments have relative higher errors compared to smaller catchments. This is because the flow varies considerably between small and large catchments. See appendix 8.3 for maps with visible catchment area.

c) Focus on average MAPE:

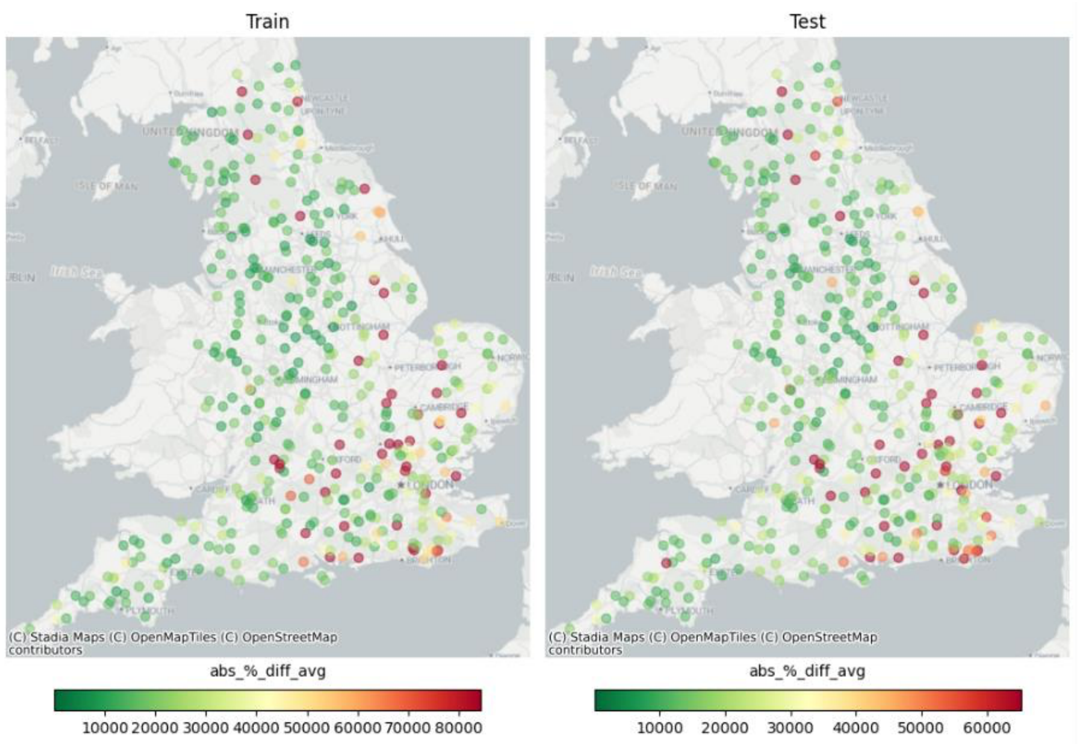


Fig. 8: Average MAPE: This map shows the average Mean Absolute Percentage Error across 409 catchments. This map visualises that although larger catchments have higher absolute errors, the errors are relatively small in their respective catchments. Whereas the errors in smaller catchments are much more significant. See appendix 8.3 for maps with visible catchment area.

d) Focus on average MAPE when excluding chalk streams:

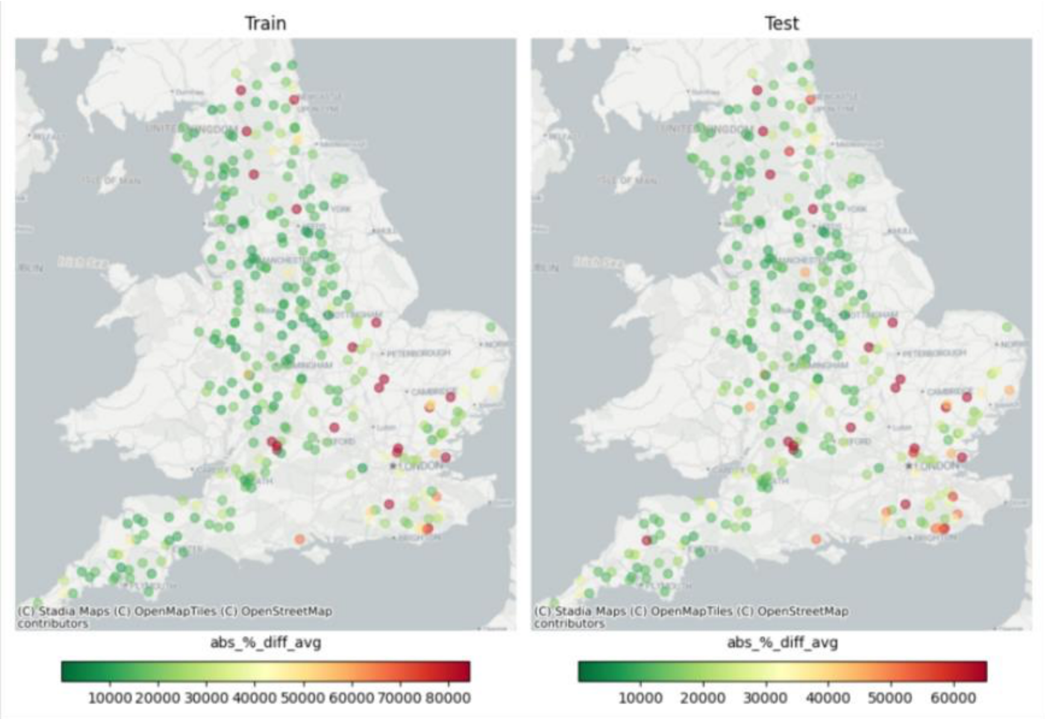


Fig. 9: No chalk streams: This map shows the mean absolute percentage error in 299 (out of 409) catchments where known chalk streams have been removed. The range of errors drops significantly compared to all catchments.

6.1.5 Examples of time series performance

Our error analysis on time series has shown good model capability for time prediction of peaks and drops, but still insufficient regarding base flow estimations, especially in conditions of low flows. Some examples below show both actual and predicted flow levels (no transformation on label applied at this stage). For each example the gauge ID, and the overall catchment Nash-Sutcliffe Efficiency (NSE) are reported in the description.

Low-performing example:

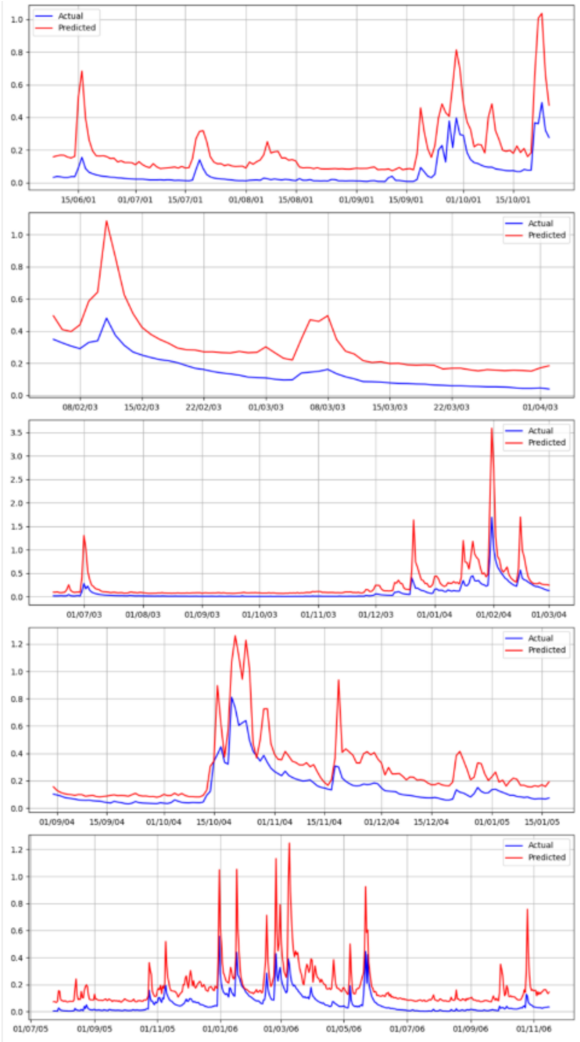


Fig. 10: Low-performing: Ancholme at Toft Newton (Gauge: 29009)
Catchment area: 27.2km2, NSE: -2.9. y-axis measured in (m^3/s).

High-performing example:

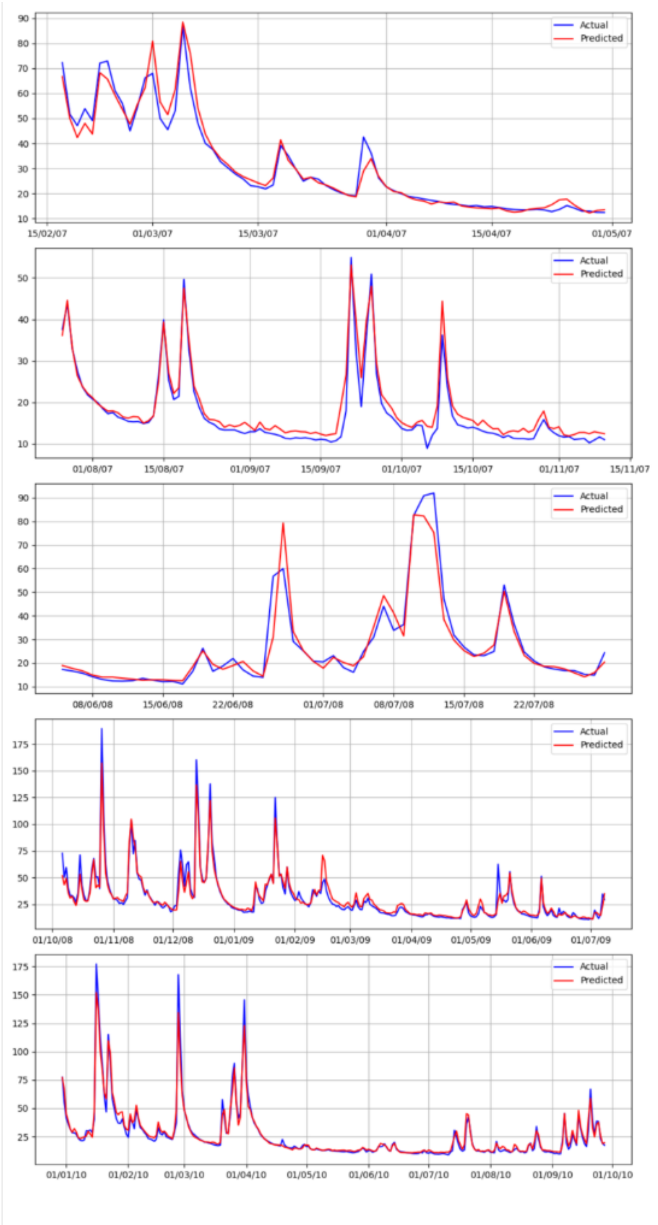


Fig. 11: **High-performing**: Aire at Beal (Gauge: 27003)
Catchment area: 1931.1km², NSE: 0.97. y-axis measured in (m^3/s).

6.1.6 Nash–Sutcliffe Efficiency coefficient (NSE)

A key metric used to evaluate hydrological models is the Nash–Sutcliffe model efficiency coefficient (NSE). NSE is useful to compare the models developed in this project with others proposed by the literature, which use NSE as a benchmark/reference statistic.

The current model achieved an average NSE of 0.59, median of 0.84, maximum of 0.97, and minimum of -19.66 (see table 4 below). Furthermore, our model exceeds 0.5 NSE for 88.9% of the 409 catchments used in the model development and test (see map of NSE below).

For comparison, [Lane et al. (2019)] found that, when tested on 1000 catchments in Great Britain, four hydrological models (TOPMODEL, ARNO, PRMS and SACRAMENTO) produced simulations that exceeded 0.5 NSE in 81%, 87%, 88% and 90% of the catchments. This suggests that the current version of our model is performing at a similar or higher level compared to traditional hydrological models. With the caveat that our model has been tested on 409 catchments, and Lane et al (2019) compared models based on 1,000 catchments.

Below is the distribution of the Nash-Sutcliffe Efficiency.:

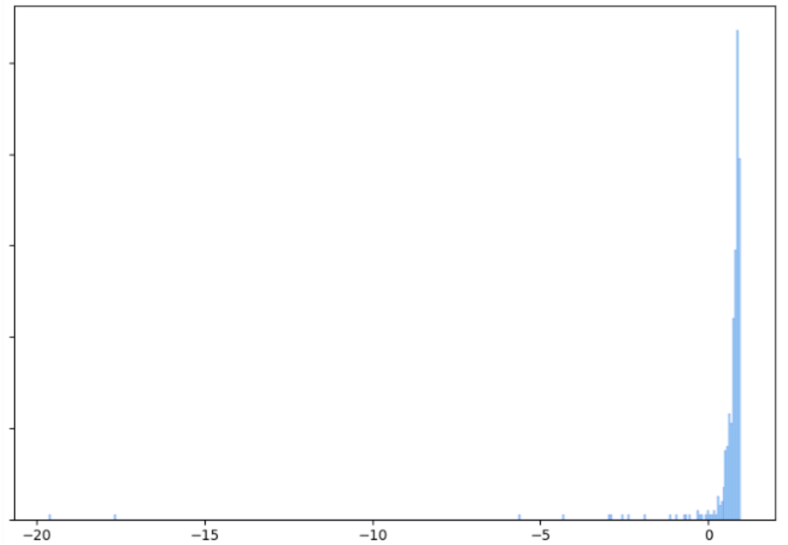


Fig. 12: Distribution of Nash-Sutcliffe Efficiency (NSE) across 409 catchments. Two poor performing catchments with -19.66 and -17.7 NSE impacts the mean NSE considerably (see table 2).

River Deep Mountain AI

Mean	Median	Max.	Min.	% Exceeding 0.5 NSE
0.56	0.84	0.97	-19.66	88.9%

Table 4: The distribution of Nash-Sutcliffe Efficiency across 409 catchments.

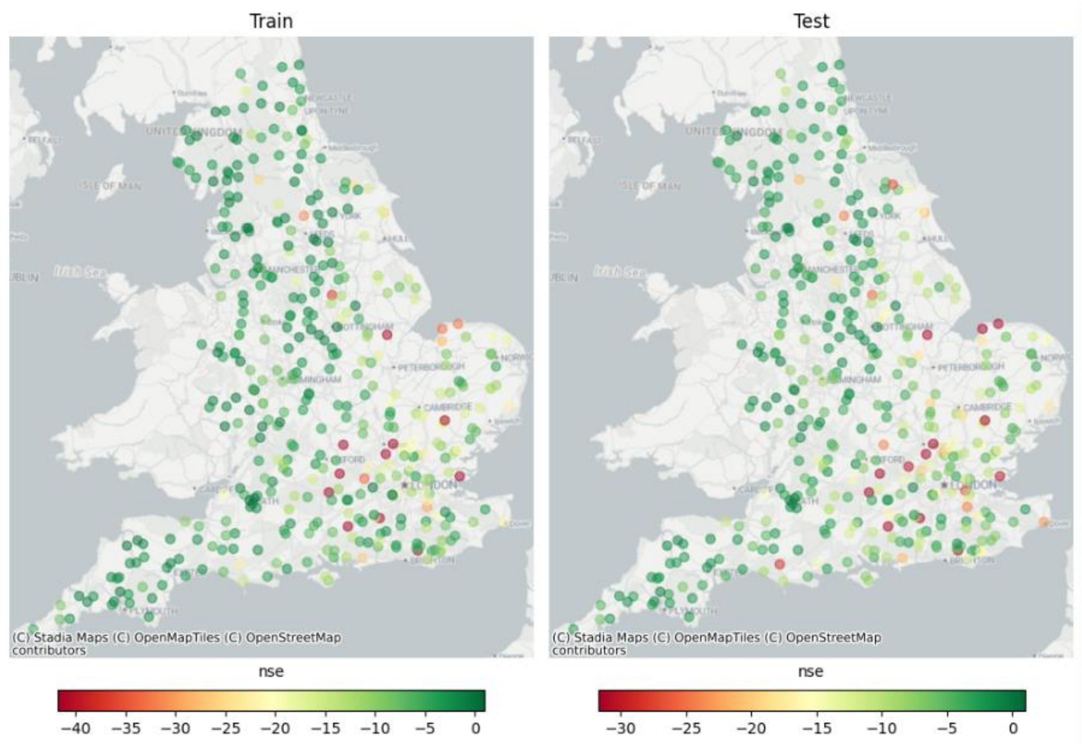


Fig. 13: **Nash-Sutcliffe Efficiency (NSE):** This map shows the distribution of NSE across 409 catchments. As highlighted above, the mean NSE is 0.56 and the median is 0.84, with 88.9% of catchments exceeding 0.5 NSE.

7 Validation

7.1 Purpose

The Open Flow Model aims to estimate daily mean river flow (m^3/s) in ungauged catchments using a machine learning approach that leverages both static and dynamic data sources. This model supports scalable and timely estimation of flow metrics, addressing data gaps in smaller or remote rivers where gauged data is unavailable.

In this section, we aim to show that we can produce a model that can generalise well to unseen catchments.

The purpose of the validation exercise is to demonstrate the robustness and generalisability of the Open Flow Model across diverse hydrological contexts. By evaluating model performance on catchments that were not included in the training set, we assess its ability to extrapolate flow predictions in ungauged regions.

7.2 Approach

For the validation, the initial plan was to use the latest version of the CAMELS GB data (2015–2025) to allow validation of model flow simulations against recent gauged flows. However, since CEH is yet to release V2 of the CAMELS GB dataset and due to time constraints on the project, we had to explore other validation approaches.

The train-test split methodology was according to a sequential time split from 1997-2015, where all NRFA gauges (and associated CAMELS GB catchments) were used in both the train and test datasets. The first 70% of time series flow for each gauge was used in the training and the remaining 30% of gauged flow was used in the testing.

The initial validation plan was to test the model on completely unseen data from the latest Camels GB dataset. However, given that CEH has not yet released V2 of the CAMELS GB data and given time constraints, validation was limited to the original data used for the training and testing. So, the validation data split needed to be different from that used in the model training and testing.

Therefore, during the validation exercise in September 2025, the same data that were used for training and testing were split geographically (using a 70%:30% split) for validation, to distinguish from the temporal split implemented for the model training and testing. This validation data split approach ensured that catchments appearing in the training set are different from those in the validation set. This approach eliminated potential model leakage (where the model inadvertently learns patterns from the validation set due to overlapping spatial or temporal characteristics with the training data, leading to overly optimistic performance estimates) and improved the model's robustness and is in line with approaches taken in similar studies.[12]

The geographic 70%:30% split does, however, introduce the risk of statistical imbalance between the train and validation sets. This is because there's a chance that the distributions of train and validate could differ when the chosen split is implemented.

Here are summary statistics for the first totally random (geographic) train validation splits:

Run	MEAN FLOW IN TRAIN (m^3/s)	MEAN FLOW IN VALIDATE (m^3/s)	STANDARD DEVIATION FLOW IN TRAIN (m^3/s)	STANDARD DEVIATION FLOW IN VALIDATE (m^3/s)
01	5.21	7.40	19.5	22.57
02	6.80	3.95	23.3	12.3
03	4.90	7.76	17.64	25.04

Table 5: Summary stats for random train validate splits.

These results confirmed that a random split was not suitable for the train validation split as a statistical imbalance was present in the target. This is evident in the standard deviation of train and test presenting unstable behavior.

To avoid this, we stratified the target to ensure the train and validation have a closer target distribution. To achieve this, we binned the target (flow) (5 bins but parameterised in the code) and used this new discrete parameter to stratify the target in the conventional way. Two sets of catchment IDs are created (train and validation). These catchments have a similar distribution of target bins in train and validate. Under this approach, a random geographic split was maintained, but stratified over target bins.

Three more distinct train validation sets were created and evaluated. This approach assessed whether the validation results were consistent across the different splits and confirmed that the model's performance was not overly sensitive to the specific composition of the training and validation data. These runs were denoted as run 04, run 05 and run 06.

Run	MEAN FLOW IN TRAIN (m^3/s)	MEAN FLOW IN VALIDATE (m^3/s)	STANDARD DEVIATION FLOW IN TRAIN (m^3/s)	STANDARD DEVIATION FLOW IN VALIDATE (m^3/s)
04	5.52	6.96	18.68	25.89
05	6.13	5.35	21.81	16.95
06	6..31	5.09	21.89	17.65

Table 6: Summary stats for random (but stratified by target) train validate splits.

7.3 Results

7.3.1 Definitions

Metric	Description
MAE (Global)	Mean Absolute Error in daily average flow calculated across all data points globally.
MEAN MAE (by catchment)	Average of the MAE values computed individually for each catchment.
MEAN NSE (by catchment)	Average of the NSE values computed individually for each catchment.
Mean (Global)	The average (mean) daily average flow calculated across all data points globally
MEAN of means (by catchment)	The mean daily average flow computed individually for each catchment.
SD (Global)	The Standard Deviation of all the data.
MEAN of SDs (by catchment)	The mean of the Standard Deviations of the catchments

7.3.2 Run 04

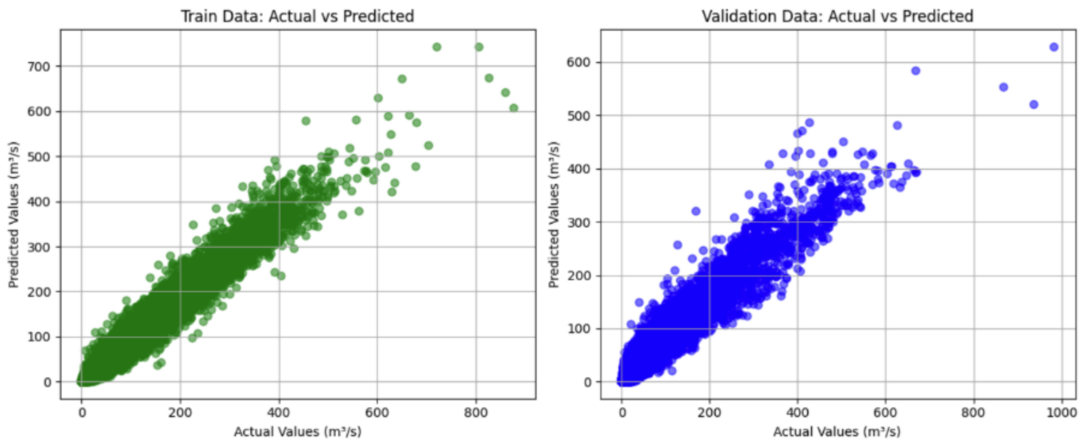


Fig. 14: Actual and predicted daily flows for train and validation for run 04

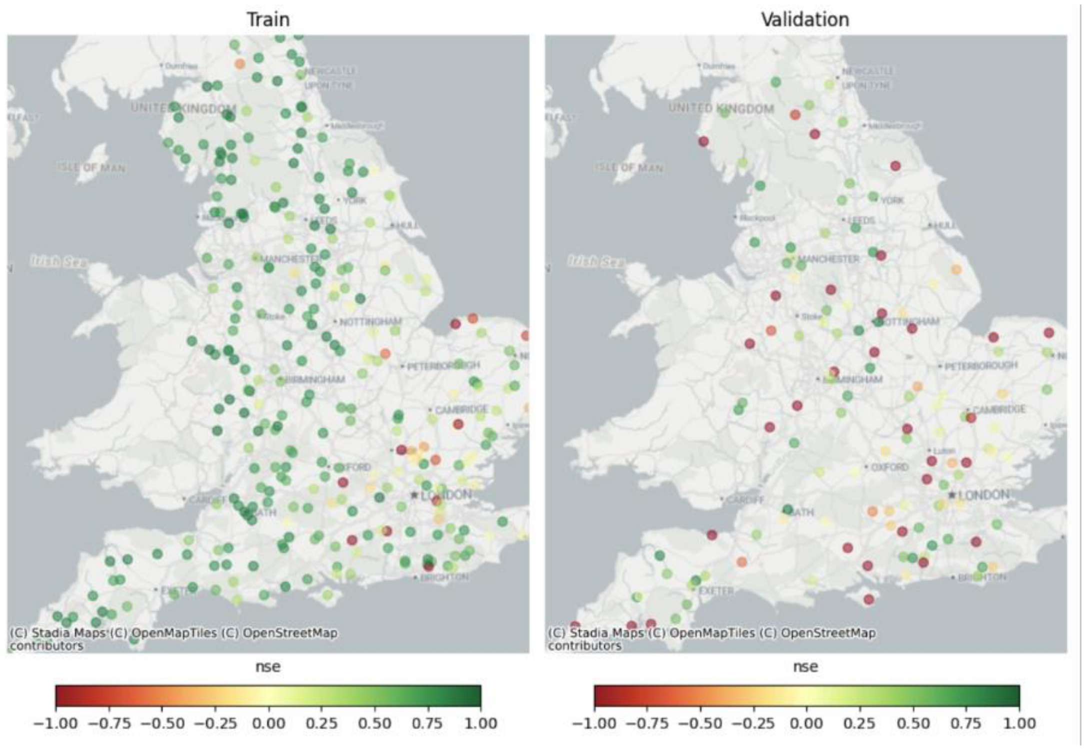


Fig. 15: Geographical representation of catchment NSE metrics

7.3.2.1 Interpretation

1. Training Data Performance
The green dots in Figure 14 are tightly clustered along the diagonal line (ideal prediction line), indicating a strong correlation between actual and predicted values. This suggests that the model has learned the training data well, with high accuracy and low error. As shown in Fig. 15, catchments with poor NSE in the training dataset appear to be concentrated in the east and southeast, possibly related to chalk stream catchments.

2. Validation Data Performance
The blue dots in Figure 14 also follow the diagonal trend but with greater dispersion, especially with higher flows. This indicates that while the model generalises reasonably well, there is more variance and less precision compared to the training set. The spread suggests potential overfitting or that the validation set contains more complex or noisy data. From a geographic perspective, as shown in Fig. 15, catchments with poor NSE were more scattered than that in the training dataset.

7.3.2.2 Results

Run	MAE (Global)	MEAN MAE (by catchment)	MEAN NSE (by catchment)	MEAN (Global)	MEAN of means (by catchment)	SD (Global)	MEAN of SDs (by catchment)
04	1.94	1.84	-0.19	6.96	6.31	25.89	7.04

Table 7: Validation run 04 results.

7.3.3 Run 05

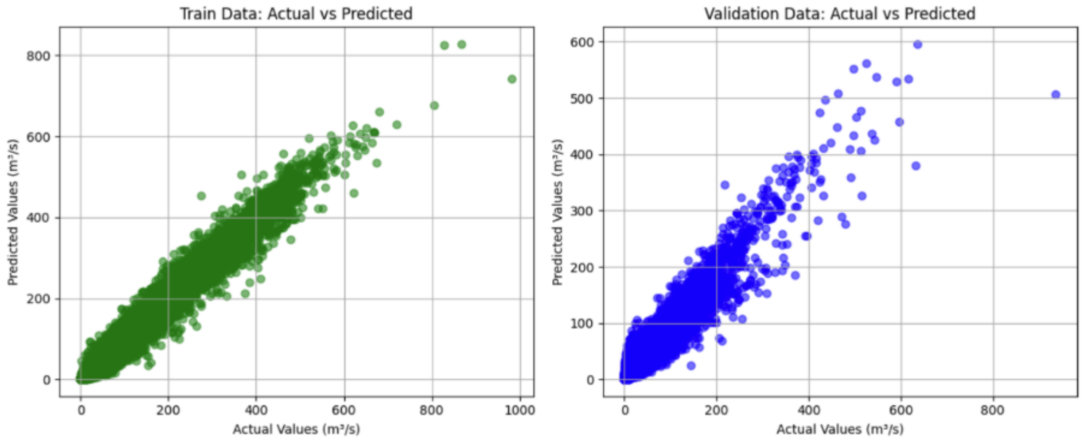


Fig. 16: Actual and predicted daily flows for train and validation for run 05

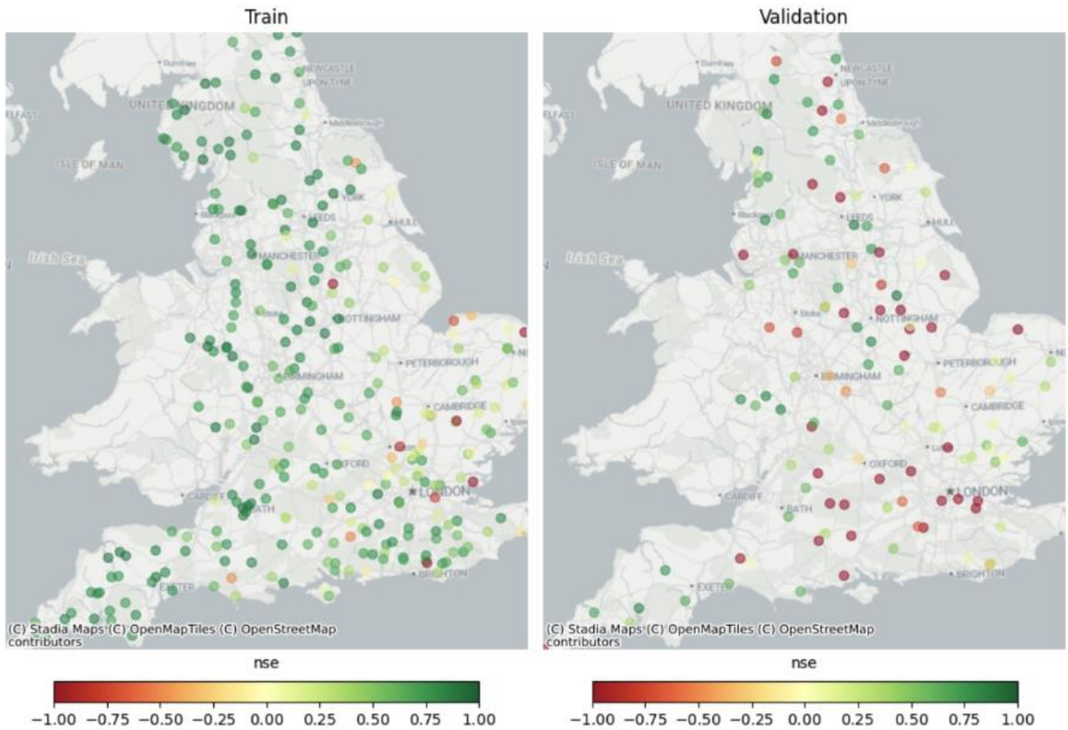


Fig. 17: Geographical representation of catchment NSE metrics

7.3.3.1 Interpretation

1. Training Data Performance
Again, the green dots are tightly clustered along the diagonal line, indicating a strong correlation between actual and predicted values. From a geographic perspective, the scatter of poor performing catchments (Fig. 17) was more dispersed than it was in run 04 (Fig. 15).

2. Validation Data Performance
Again, the blue dots also follow the diagonal trend but with greater dispersion, especially with higher flows. As was the case in run 04, catchments with poor NSE performance are widely scattered (Fig. 17), with no clear geographic pattern.

7.3.3.2 Results

Run	MAE (Global)	MEAN MAE (by catchment)	MEAN NSE (by catchment)	MEAN (Global)	MEAN of means (by catchment)	SD (Global)	MEAN of SDs (by catchment)
05	1.57	1.43	-6.74	5.35	4.8	16.95	5.54

Table 8: Validation run 05 results.

7.3.4 Run 06

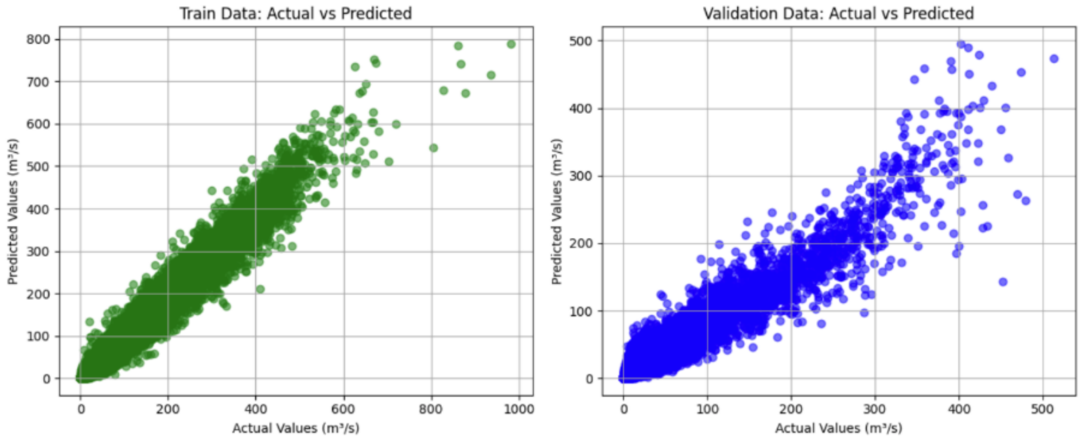


Fig. 18: Actual vs predicted for train and validation for run 06

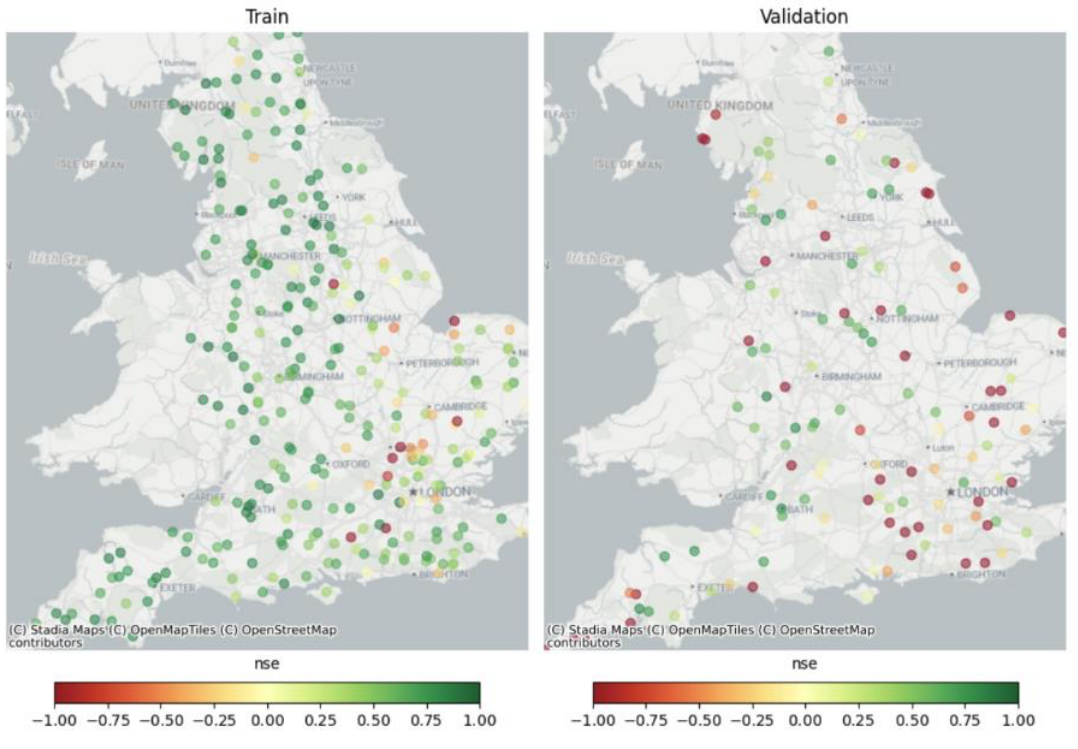


Fig. 19: Geographical representation of catchments NSE metrics

7.3.4.1 Interpretation

1. Training Data Performance
Again, the diagonal alignment of points suggests a strong correlation between actual and predicted values. As was the case in run 04 (Fig. 15), poor performing catchments appear to be concentrated in the east and southeast (Fig. 19).

2. Validation Data Performance
While the model demonstrates strong predictive alignment across most of the dataset, it consistently fails to capture very high actual values in runs 04, 05, and 06, instead predicting significantly lower outputs. This systematic underestimation indicates a limitation in the model's ability to generalise to outlier scenarios, which may compromise its reliability in high-stakes or edge-case applications. Consistent with the previous runs, poor performing catchments in the validation dataset appear to be widely scattered, with no clear geographic patterns (Fig. 19).

7.3.5.2 Results

Run	MAE (Global)	MEAN MAE (by catchment)	MEAN NSE (by catchment)	MEAN (Global)	MEAN of means (by catchment)	SD (Global)	MEAN of SDs (by catchment)

06	1.6	1.59	-0.75	5.09	5.07	17.65	6.06

Table 9: Validation run 06 results.

7.3.6 Results overview

The following table summarises the model performance across the three runs:

Run	MAE (Global)	MEAN MAE (by catchment)	MEAN NSE (by catchment)	MEAN (Global)	MEAN of means (by catchment)	SD (Global)	MEAN of SDs (by catchment)
04	1.94	1.84	-0.19	6.96	6.31	25.89	7.04
05	1.57	1.43	-6.74	5.35	4.8	16.95	5.54
06	1.6	1.59	-0.75	5.09	5.07	17.65	6.06

Table 10: Overview table of validation runs.

The models demonstrate consistent performance across all runs, especially with respect to MAE. This shows that there is no significant statistical differences between the train and validation datasets for all runs. One marginal exception worth nothing is the increased MAE in run 04. This was expected, however, due to the high mean of the target in test. See table 6. This further highlights the need for carefully prepared train-validation splits.

From this, we can conclude that there was sufficiently little bias in the selection of the train-validation splits.

In terms of accuracy, however, the *MEAN NSE (by catchment)* scores of the validation runs don't show the entire picture. Therefore, a more detailed analysis of the results was conducted. For this purpose, we broke the predictions down by catchment and grouped higher and lower performing catchments. The aim was to split the data into two groups (higher and lower performing catchments). This would give us an overview of the results distribution and pave the way for further analysis.

The results of these breakdowns are shown in the tables below.

Run	MEAN MAE (by high-performi	MAE (max high-performing	MAE (min high-performing	Coverage (%)	MEAN MAE (by low-performing	MAE (max low-performing catchments)	MAE (min low-performing	Coverage (%)

	ng catchments)	catchments)	catchments)		catchments)		catchments)	
04	0.63	2.8	0.04	79.67	6.85	27.19	2.94	20.33
05	0.73	2.97	0.03	79.83	4.65	10.24	3.04	20.17
06	0.72	2.36	0.06	79.99	5.16	14.51	2.37	20.01

Table 11: MAE Results on high and low-performing catchments

Run	MEAN NSE (by high-performing catchments)	NSE (max high-performing catchments)	NSE (min high-performing catchments)	Coverage (%)	MEAN NSE (by low-performing catchments)	NSE (max low-performing catchments)	NSE (min low-performing catchments)	Coverage (%)
04	0.62	0.95	-0.1	79.28	-3.4	-0.16	-17.53	20.72
05	0.59	0.97	-0.39	79.25	-36.98	-0.79	-636.58	20.75
06	0.54	0.96	-0.5	80.04	-5.81	-0.61	-49.36	19.96

Table 12: NSE Results on high and low-performing catchments

The model demonstrates strong and consistent performance across the most accurate catchments (with respect to MAE and NSE), which account for approximately 80% of the dataset. Further investigation could include looking at catchment size and average flow among the high and low performing catchments. Patterns in the geographical distributions of poor performing catchments can also be related to regional catchment characteristics, such as chalk streams.

As part of the analysis, we also created a series of linear correlation matrixes for each run (see section 9.4). These plots highlighted that the target was highly related to the high error. Further analysis is recommended.

In addition to the above analysis, more details of plots of the NSE scores were created and can be found in section 9.5 of the appendix.

We also noticed that, interestingly, in some catchments where the MAE is higher, the NSE improves (see Figures 20-22). This suggests that while the predictions may deviate in magnitude, they still follow the correct trend and variability of the observed data.

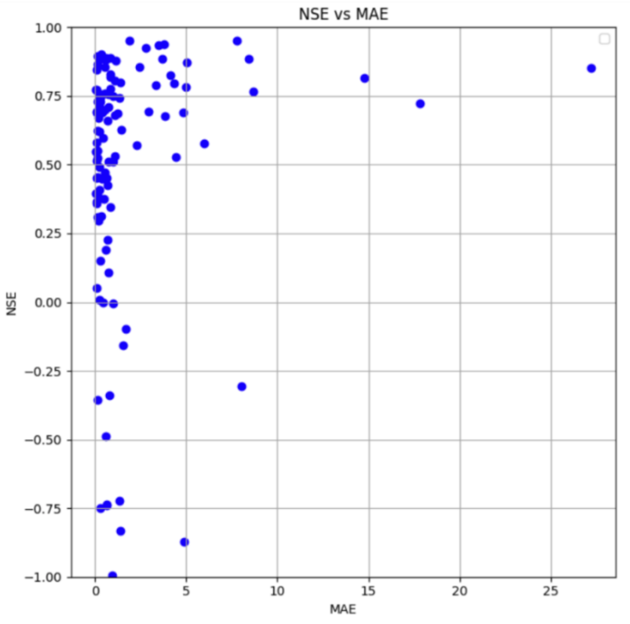


Fig. 20: MAE vs NSE for run 04 (data only shown where NSE is greater than −1).

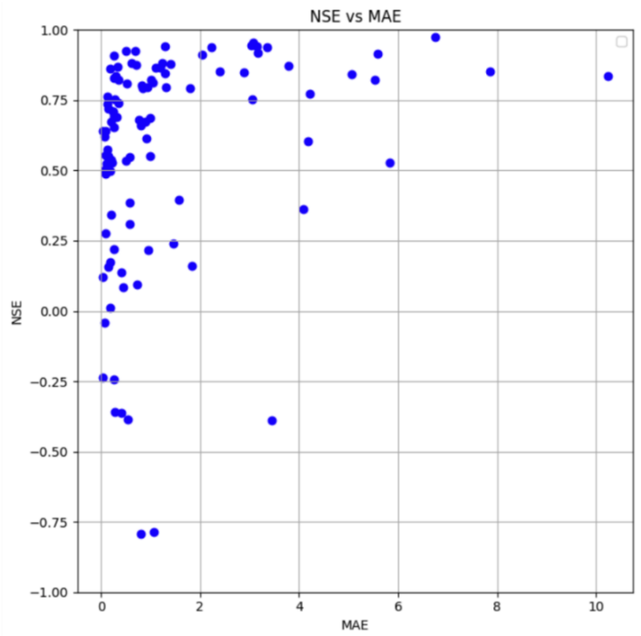


Fig. 21: MAE vs NSE for run 05 (data only shown where NSE is greater than −1).

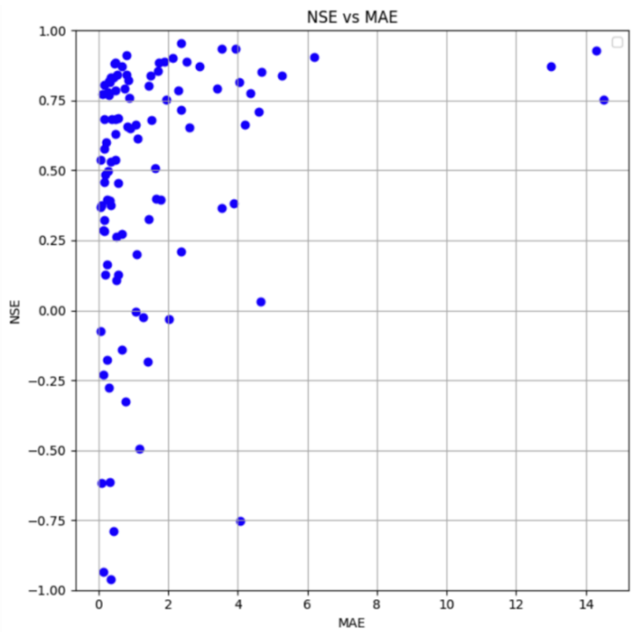


Fig. 22: MAE vs NSE for run 06 (data only shown where NSE is greater than −1).

This pattern indicates that the model retains predictive value, even in less accurate regions, though the scale of predictions may require adjustment. We recommend conducting further investigations to understand the underlying factors contributing to this behaviour and to explore potential calibration strategies. Some initial thoughts could be that there's a higher proportion of chalk stream catchments in the low-performing catchments, as we know they're notoriously difficult to model. For further evidence of this please see figures 23-28 in the appendix.

7.4 Summary

The Open Flow Model demonstrates strong generalisation across unseen catchments, with consistent performance across multiple validation runs. While some catchments remain challenging, the model reliably estimates river flow for the majority of cases. The revised strategy for model validation – especially the exclusive train-validation split and multi-run stratified evaluation – was key to ensuring robustness and credibility.

The validation results were expected given the test results. The mean MAE for the test results was 1.015 compared to the mean results of the 3 validation runs of 1.7033. The loss in performance is probably due to the leakage during the train test split made in validation.

8 Final considerations and next actions

From the error analysis, it is clear the model is underperforming the most in a relatively small set of catchments. The research is currently focused on understanding the motivations for this behaviour, which does not seem entirely linked to a single or a small set of attributes.

There are some recommended improvements, which are in order of importance:

1. Applying log transformation to simulated flow to shrink the distribution, thereby helping the algorithm identify the optimal solution.

2. Re-evaluate the link between Natural England's chalk stream dataset and catchments used in this model.

3. Adding a further set of statistics, potentially more on observation-level, to help SMEs in their investigation.

4. Continue investigation of the poorest performing catchments, for two reasons:
   a. To understand what might be done to improve model performance in these cases. See section 9.4.
   b. To help identify which catchments the model is suitable for and not.

5. Running hyperparameter optimisation research to potentially reduce the dimension

6. Further investigation into unusual finding when looking at high MAE (large error) and higher performing NSE catchments during the validation exercise.
7. In validation, even the stratified split of the target still shows signs of some statistical imbalance. In validation we used binning with 5 bins. Other techniques for splitting could be explored including testing more bins.

9 Appendix

9.1 Attribute features list

Category	Field
Hydrologic Attributes	baseflow_index
Soil Attributes	sand_perc
	silt_perc
	clay_perc
	organic_perc
Topographic Attributes	gauge_elev
	area
	dpsbar
	elev_mean
	elev_min
	elev_10
	elev_50
	elev_90
	elev_max
Landcover Attributes	dwood_perc
	ewood_perc
	grass_perc
	shrub_perc
	crop_perc
	urban_perc
	inwater_perc
	bares_perc
Human Influence Attributes	surfacewater_abs
	groundwater_abs
	discharges
	num_reservoir
	reservoir_cap
Chalk Streams	chalk_stream_flag

9.2 Timeseries features list

Category	Field
Weather	precipitation
	temperature
	humidity
	shortwave_rad
	longwave_rad
	windspeed
Available Labels	discharge_vol
	log1p_discharge_vol
Time Reference	sin_year
	cos_year
	time_ref

9.3 Timeseries plots for validation exercise

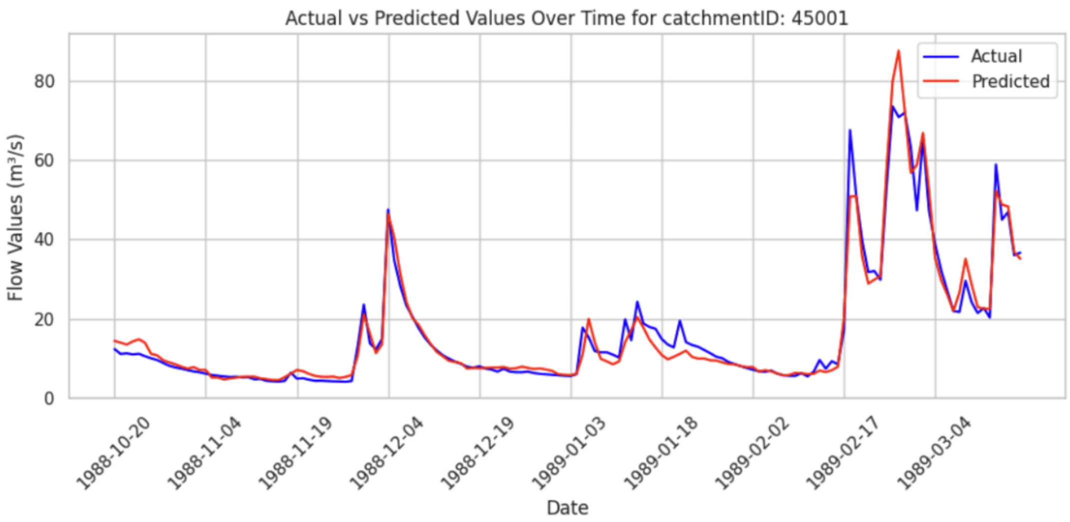


Fig. 23: Timeseries plots for actual vs predicted for high-performing catchment in run 04

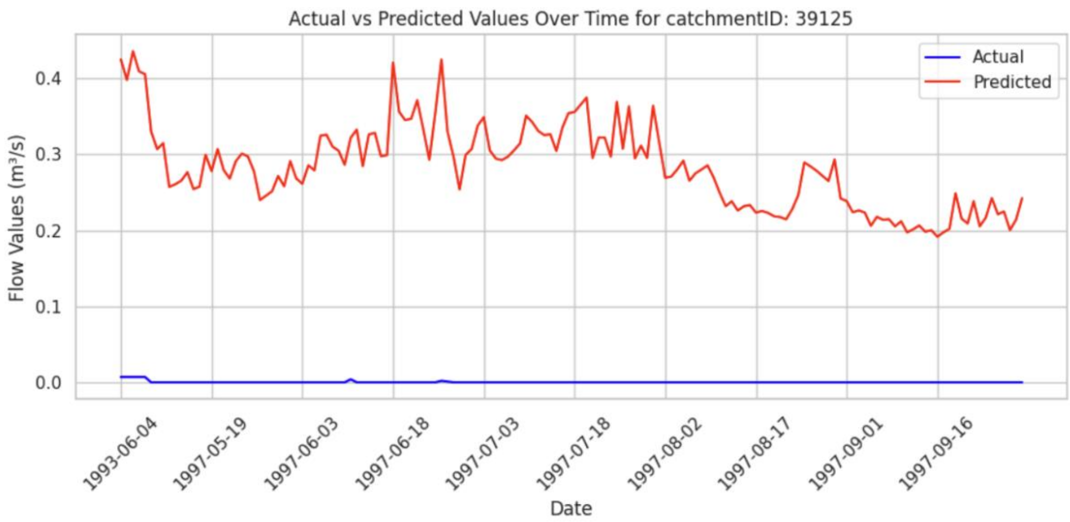


Fig. 24: Timeseries plots for actual vs predicted for low-performing catchment in run 04

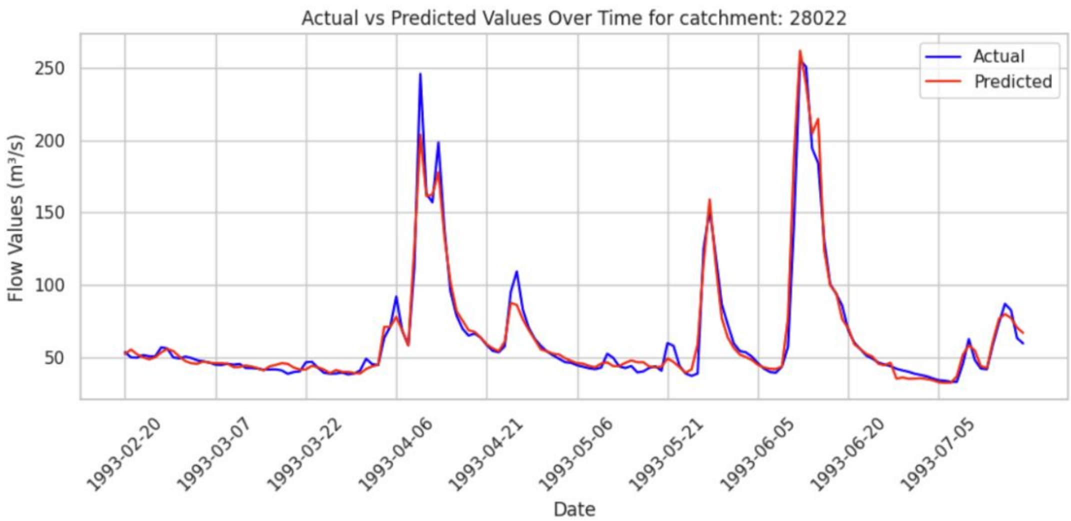


Fig. 25: Timeseries plots for actual vs predicted for high-performing catchment in run 05

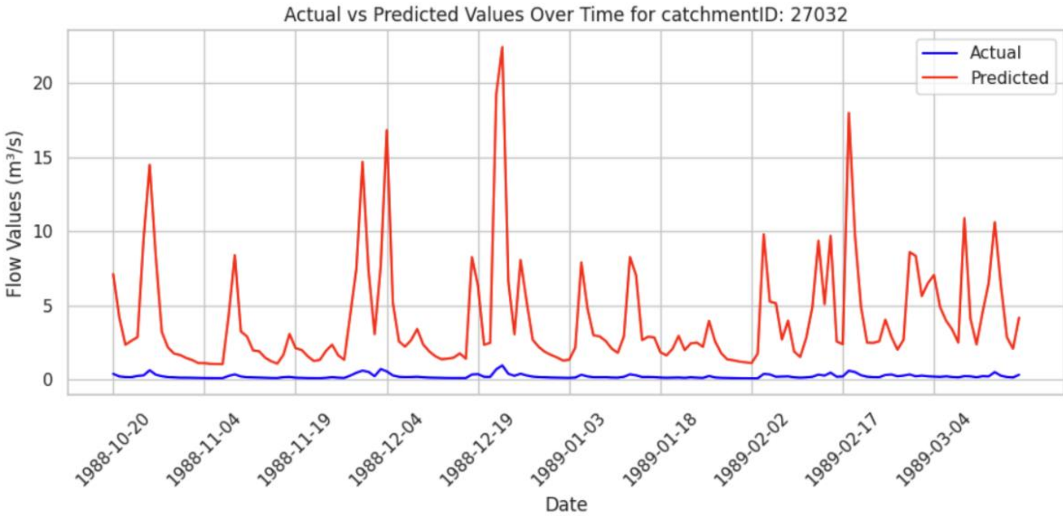


Fig. 26: Timeseries plots for actual vs predicted for low-performing catchment in run 05

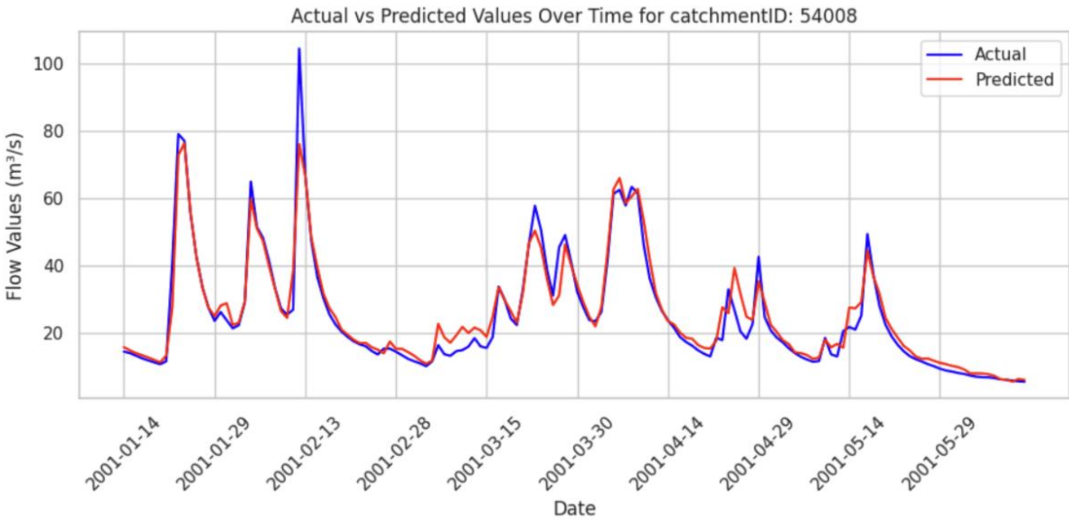


Fig. 27: Timeseries plots for actual vs predicted for high-performing catchment in run 06

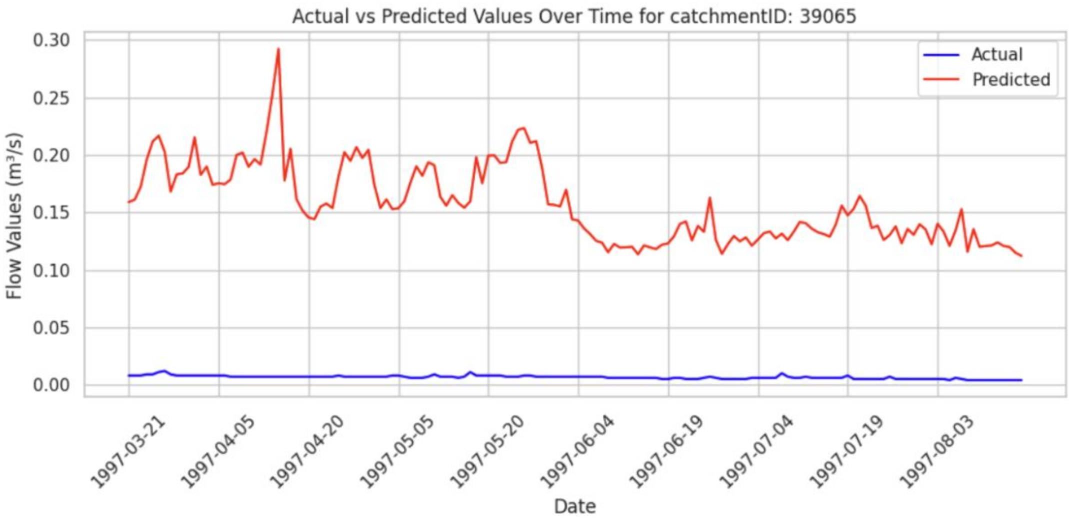


Fig. 28: Timeseries plots for actual vs predicted for low-performing catchment in run 06

9.4 Maps with catchment area

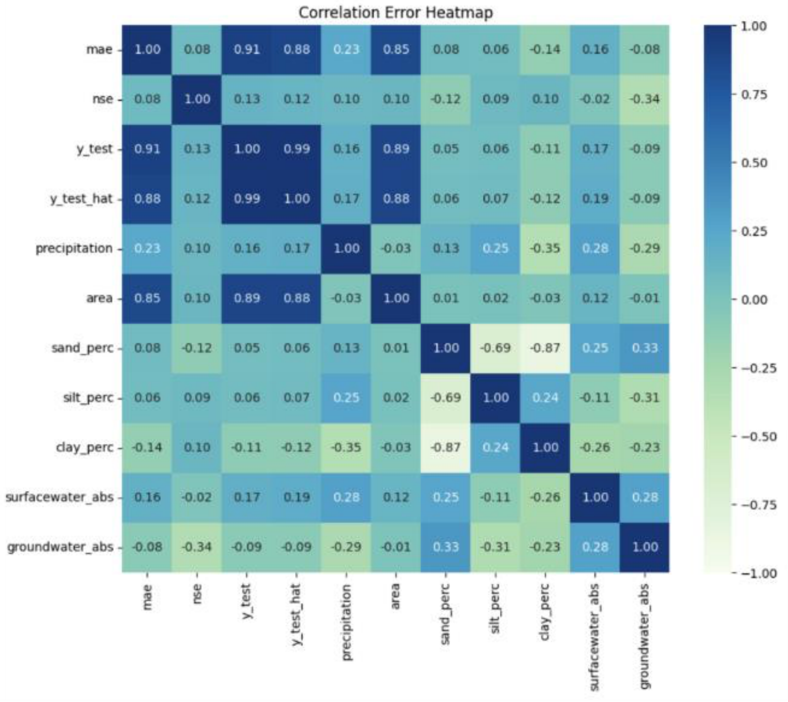


Fig. 29: Correlation plot for test run 04 results.

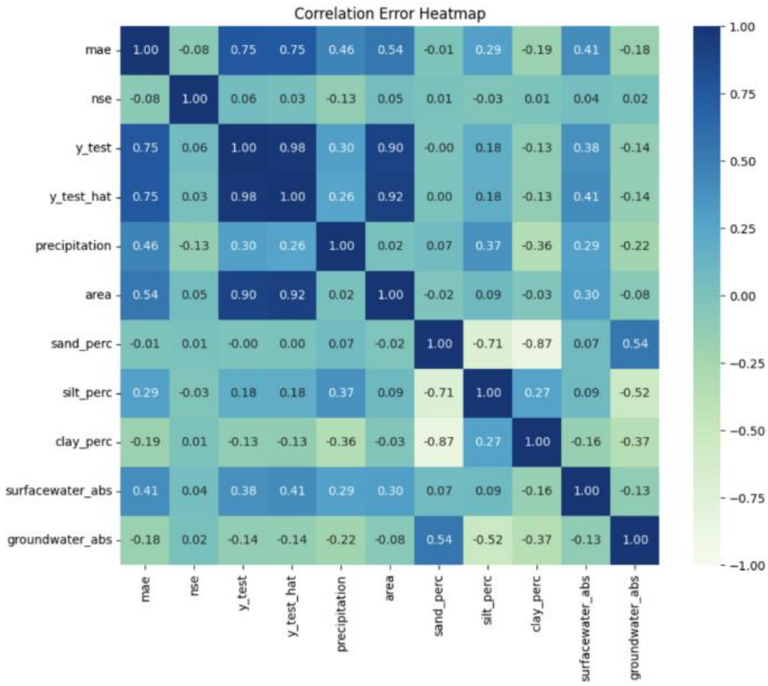


Fig. 30: Correlation plot for test run 05 results.

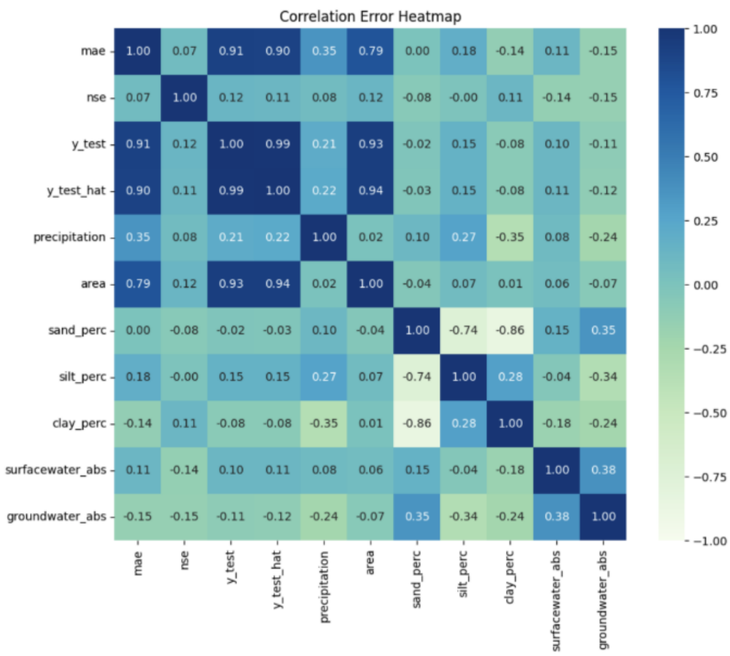


Fig. 31: Correlation plot for test run 06 results.

9.5 Validation NSE distributions for runs 04, 05 and 06.

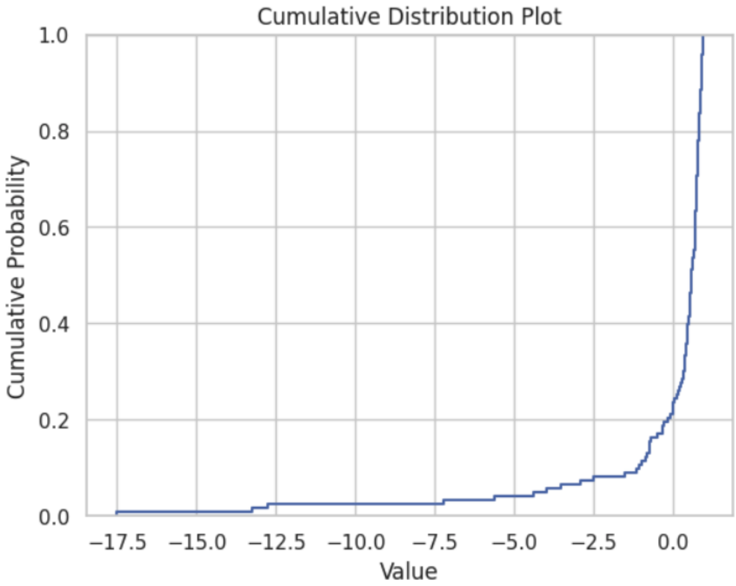


Fig. 32: Cumulative Distribution plot for NSE (run 04).

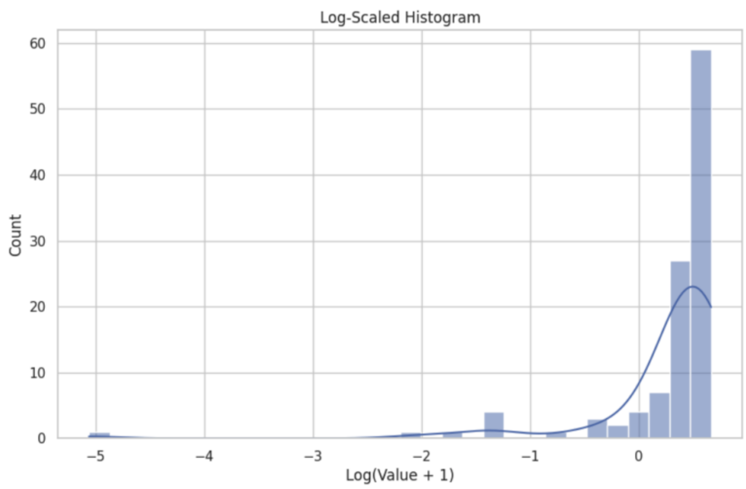


Fig. 33: Log scaled histogram plot for NSE (run 04).

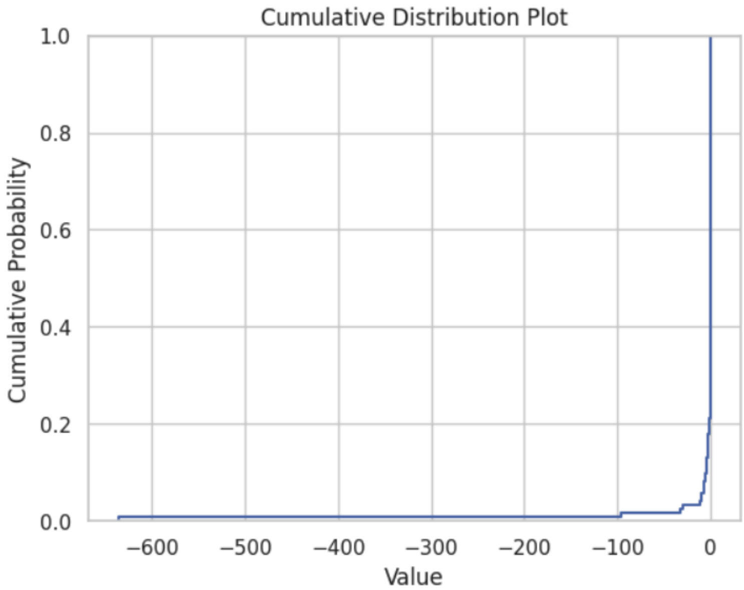


Fig. 34: Cumulative Distribution plot for NSE (run 05).

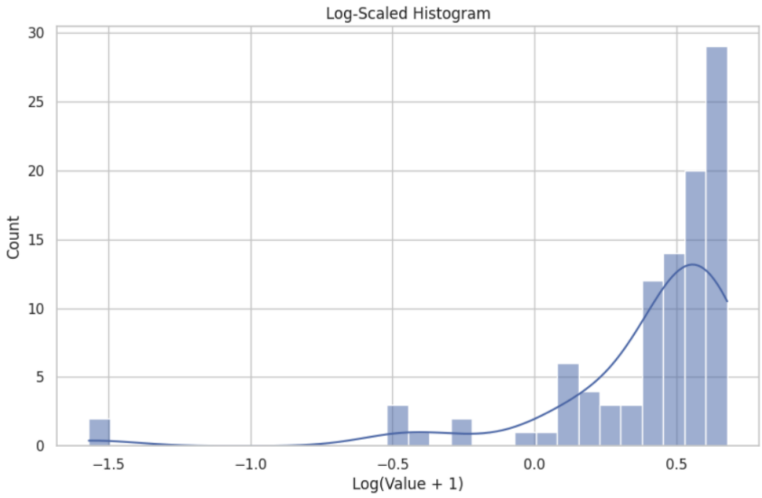


Fig. 35: Log scaled histogram plot for NSE (run 05).

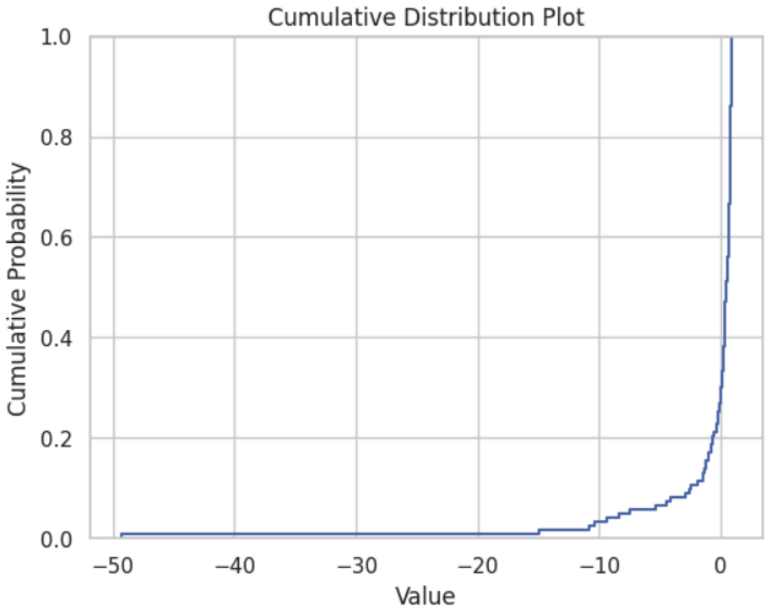


Fig. 36: Cumulative Distribution plot for NSE (run 06).

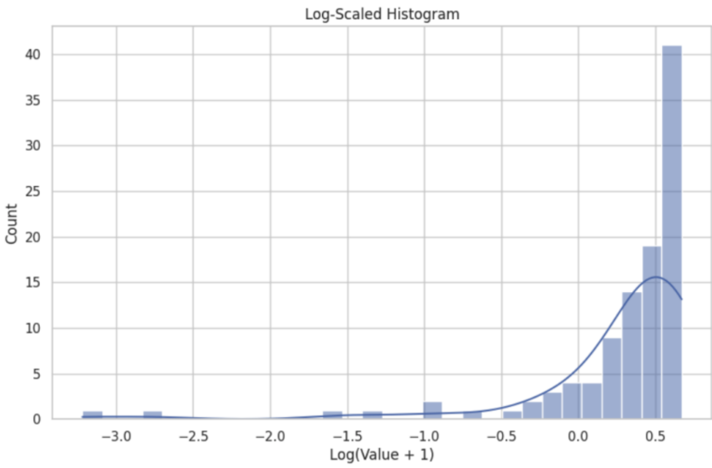


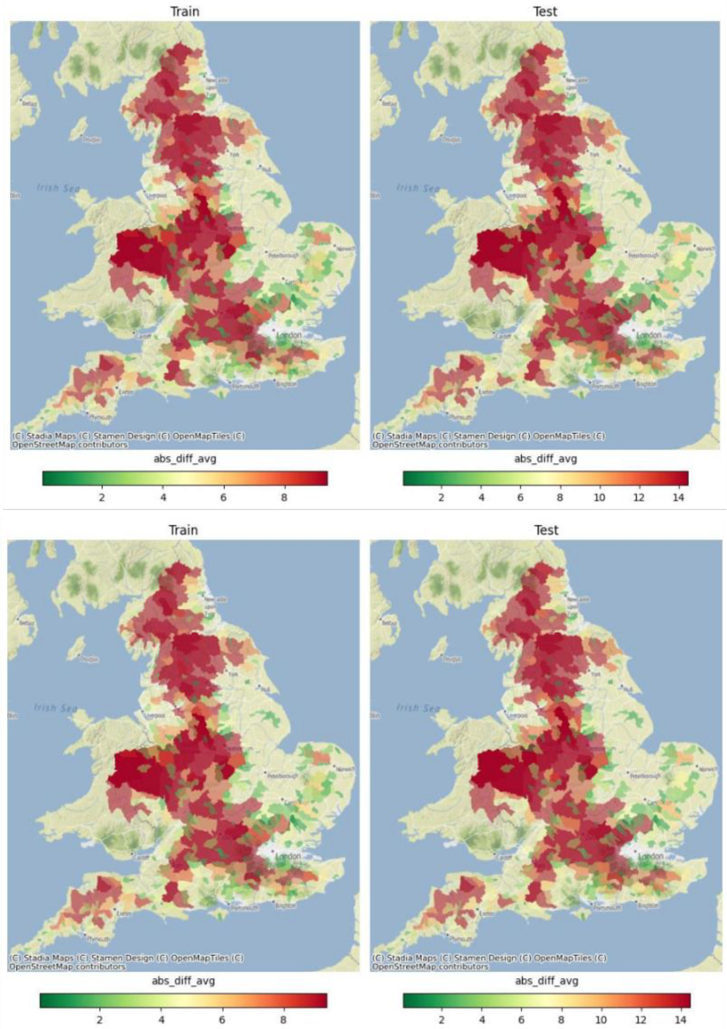
Fig. 37: Log scaled histogram plot for NSE (run 06).

Run	MEAN	MEDIAN	MAX	MIN	Percentage of catchments ≥ 0.5 NSE (%)
04	-0.19	0.58	0.95	-17.53	58.5
05	-6.74	0.6	0.97	-636.58	59.3
06	-0.75	0.5	0.96	-49.36	49.6

Table. 13: Detailed NSE metrics for validation runs.

9.6 Maps with catchment area

A) MAE with catchment area



B) MAPE with catchment area